

SPATIAL DYNAMICS OF AN INTEGRO-DIFFERENCE LOTKA-VOLTERRA MODEL IN A SHIFTING HABITAT

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SUMMARY

We study a two-species integro-difference competition model in a shifting habitat, where nonlocal dispersal is coupled with Lotka–Volterra type interactions and a favorable environment moving at constant speed. Using comparison principles and linear determinacy, we characterize asymptotic spreading speeds for the associated single-species models and identify critical thresholds for habitat tracking.

We prove that both species go extinct if the habitat shifts faster than their spreading capabilities, while competitive exclusion occurs when the shifting speed lies between the two single-species spreading speeds. For sufficiently slow habitat shift, we show that weak competition leads to coexistence and joint spreading, with solutions converging locally to the positive coexistence equilibrium in an expanding region. Under strong competition, the faster species excludes its competitor even when both can persist in isolation. These results extend reaction–diffusion theories to integro-difference systems and clarify the role of nonlocal dispersal in persistence under shifting environments. In addition, we propose a physics-informed neural network (PINN) framework to approximate traveling wave solutions and the associated wave speed, treating the wave speed as a free parameter within the network. This computational approach complements the analytical results and provides a flexible tool for investigating wave profiles and spreading dynamics in nonlocal, shifting environments.

Key words: *Nonlocal dispersal, Integro-difference model, Spreading Speeds, Lotka-Volterra model, Physics-informed Neural Network.*

1 INTRODUCTION

Climate-driven habitat shifts pose significant challenges to the persistence of spatially distributed populations [1, 2]. When favorable environments move through space at a constant speed, species must track the shifting habitat through dispersal and growth or face extinction. Mathematical models play an essential role in identifying the mechanisms governing persistence, spreading, and competitive outcomes under such dynamics [3, 4, 5].

While reaction–diffusion models have been extensively used to study population dynamics in shifting habitats, many species reproduce in discrete generations and disperse nonlocally, making integro-difference equations more appropriate [6, 7, 8]. Nonlocal dispersal can substantially alter spreading behavior and competitive interactions, yet multi-species integro-difference models in shifting environments remain less understood.

In this study, we study a two-species integro-difference competition model in a shifting habitat with Lotka–Volterra type interactions. We analyze how habitat shift speed, dispersal kernels, and competition strength determine extinction, competitive exclusion, and coexistence. By establishing spreading speed thresholds for the associated single-species models, we extend shifting-habitat theory to integro-difference systems and provide new insights into persistence under climate-induced range shifts.

We consider the following two-species competition model in a shifting habitat

$$\begin{aligned} u_{n+1}(x) &= \int_{-\infty}^{\infty} k_1(x-y) \frac{(1 + \rho_1(y-cn))u_n(y)}{1 + \rho_1(y-cn)(u_n(y) + \alpha_1 v_n(y))} dy, \\ v_{n+1}(x) &= \int_{-\infty}^{\infty} k_2(x-y) \frac{(1 + \rho_1(y-cn))v_n(y)}{1 + \rho_1(y-cn)(v_n(y) + \alpha_2 u_n(y))} dy. \end{aligned} \quad (1.1)$$

Throughout this paper we assume the following hypothesis:

- (i) $\rho_1(x)$ is non-decreasing in x for $-\infty < x < \infty$. $\rho_1(\infty) = \lim_{x \rightarrow \infty} \rho(x)$ and $\rho_1(-\infty) = \lim_{x \rightarrow -\infty} \rho(x)$ are finite, $-1 < \rho_1(-\infty) < 0 < \rho_1(\infty)$. If we consider this case, we need the condition $1 - u_n - \alpha_1 v_n \geq 0$ and $1 - v_n - \alpha_2 u_n \geq 0$ since we want the functions inside the integrals are nonnegative.
- (ii) For $i = 1, 2$, $k_i(x)$ is a bounded, nonnegative and continuous probability distribution with $\int_{-\infty}^{\infty} k_i(x) dx = 1$, and the moment generating function $\bar{k}_i(\mu) := \int_{-\infty}^{\infty} k_i(x) e^{\mu x} dx$ is convergent for one positive value and one negative value of μ .

Let

$$\begin{aligned} c_1^* &:= \inf_{\mu > 0} \frac{\ln((1 + \rho_1(\infty))\bar{k}_1(\mu))}{\mu} \\ c_{1-}^* &:= \inf_{\mu > 0} \frac{\ln((1 + \rho_1(\infty))\bar{k}_1(-\mu))}{\mu} \\ c_2^* &:= \inf_{\mu > 0} \frac{\ln((1 + \rho_1(\infty))\bar{k}_2(\mu))}{\mu} \\ c_{2-}^* &:= \inf_{\mu > 0} \frac{\ln((1 + \rho_1(\infty))\bar{k}_2(-\mu))}{\mu} \end{aligned}$$

2 MAIN THEOREMS

We propose to establish the following three theorems:

Theorem 2.1 *Suppose the initial function $(u_0(x), v_0(x))$ is nonnegative, bounded and has a nonempty compact support. If $c > \max\{c_1^*, c_2^*\}$, then*

$$\lim_{n \rightarrow \infty} (u_n(x), v_n(x)) = (0, 0)$$

uniformly for $x \in \mathbb{R}$.

Theorem 2.2 *Suppose the initial function $(u_0(x), v_0(x))$ is nonnegative, bounded, and has a nonempty compact support. If*

$$c_1^* < c < c_2^*,$$

then

$$\lim_{n \rightarrow \infty} u_n(x) = 0 \quad \text{uniformly for } x \in \mathbb{R},$$

and there exists $\eta > 0$ such that

$$\liminf_{n \rightarrow \infty} \inf_{|x-cn| \leq \eta n} v_n(x) > 0.$$

Moreover, $v_n(x)$ spreads asymptotically with speed c_2^ .*

Theorem 2.3 Suppose the initial function $(u_0(x), v_0(x))$ is nonnegative, bounded, and satisfies

$$u_0(x) \not\equiv 0, \quad v_0(x) \not\equiv 0,$$

with compact support. Assume

$$0 < \alpha_1 < 1, \quad 0 < \alpha_2 < 1,$$

and

$$0 < c < \min\{c_1^*, c_2^*\}.$$

Then both species persist and spread. More precisely, there exists $c^\dagger > c$ such that for any $0 < c' < c^\dagger$,

$$\lim_{n \rightarrow \infty} \sup_{|x - c'n| \leq R} |(u_n(x), v_n(x)) - (u^*, v^*)| = 0$$

for every $R > 0$, where (u^*, v^*) is the positive coexistence equilibrium of the corresponding Lotka–Volterra system.

We propose developing a physics-informed neural network (PINN) framework to approximate both the traveling wave solution and the associated wave speed, with the wave speed treated as an additional free parameter[9, 10]. Under suitable assumptions, one expects that as the training loss approaches zero, the neural network approximation will converge to the analytic traveling wave profile, while the free parameter consistently recovers the true wave speed for the Lotka–Volterra model.

3 RESULTS AND CONCLUSIONS

We established threshold conditions for extinction, competitive exclusion, and coexistence in a two-species integro-difference competition model with a shifting habitat. If the habitat shifts faster than the spreading capabilities of both species, then both populations go extinct. When the shifting speed lies between the two single-species spreading speeds, the faster species persists and spreads while the slower species becomes extinct. For sufficiently slow habitat shift, weak interspecific competition leads to coexistence and joint spreading, with solutions converging locally to the positive coexistence equilibrium behind the invasion front. Under strong competition, the faster species can exclude its competitor even when both species can persist in isolation. These results extend shifting-habitat theory to integro-difference systems and highlight the impact of nonlocal dispersal on persistence under environmental change.

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